

6	(a)	The magnetic flux ϕ is defined as the product of the area and the magnetic flux density that is perpendicular to it.		[B1]
	(b)	(i)	$\Phi_N = NBA \sin \theta$ $= 650 (2.5 \times 10^{-3}) (20)(35)(10^{-3})^2 \sin 30^\circ$ $= 5.69 \times 10^{-4} \text{ wb turns}$	[M1] [M1] [A1]
		(ii)	$E = N \Delta \Phi / t$ $= N (\Phi_f - \Phi_i) / t$ $= 0 - 5.69 \times 10^{-4} / (5 \times 10^{-3})$ $= 0.114 \text{ V}$	[M1] [A1]
		(iv)	$0.114 = I(5)$ $I = 0.0228 \text{ A}$ $Q = It = 0.0228 \times 5 \times 10^{-3}$ $= 114 \times 10^{-6} \text{ C}$	[C1] [A1]